

Inflationary Impact on Retail Consumer Basket

Composition

Problem Statement: 7

Domain: Finance

Team No. 07

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IDEA TITLE

Core Problem:

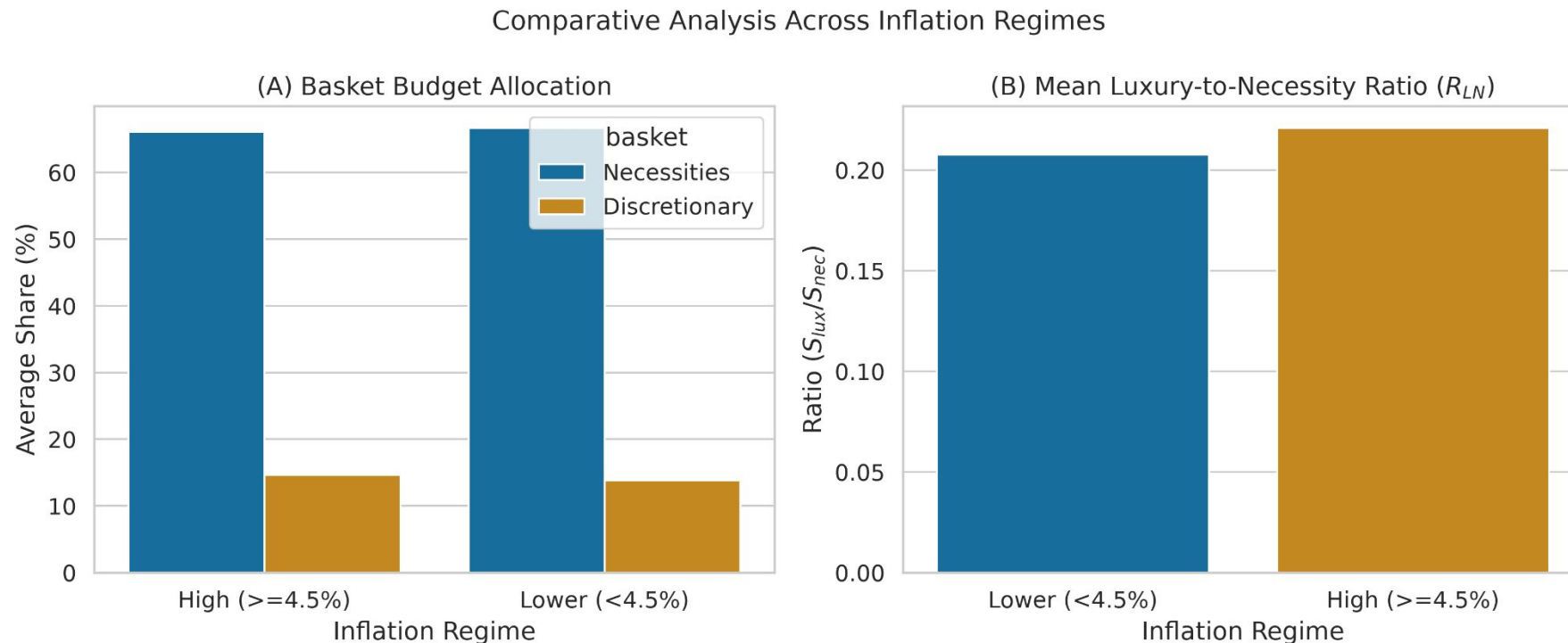
Inflation alters consumer spending dynamically, forcing shifts from luxury goods to essential staples, which traditional static retail models fail to capture.

Proposed Solution:

- A Python-based analysis tool that tracks how customer spending baskets changed between 2020 and 2023.
- Uses data thresholds to find the exact inflation rate where buying patterns start to flip.

Key Innovations & Differentiators:

- Compares low-inflation vs. high-inflation periods directly instead of treating all data the same.
- Tracks how spending shifts across categories as a whole, not just single product prices.

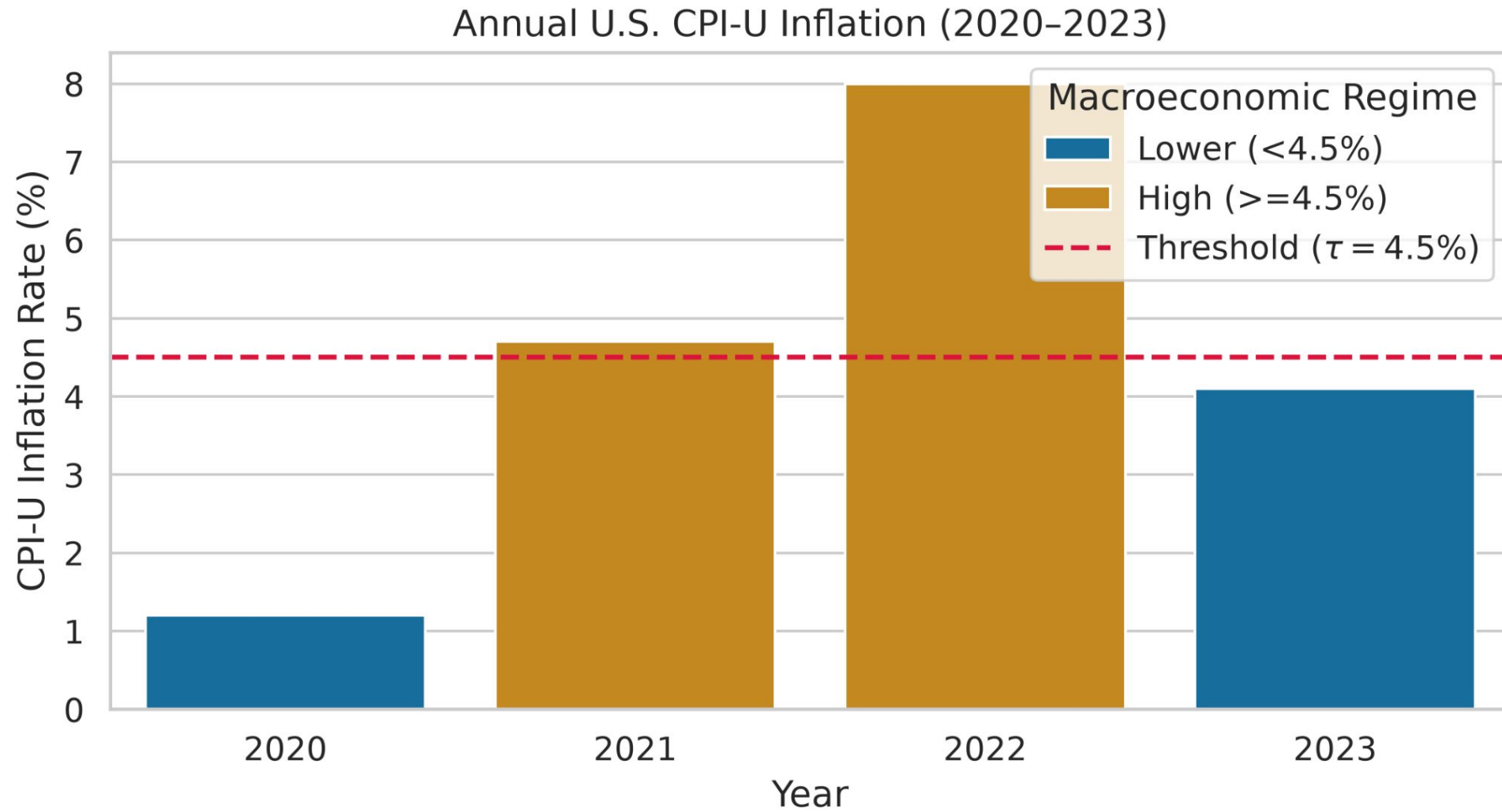


TECHNICAL APPROACH

Tech Stack: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook.

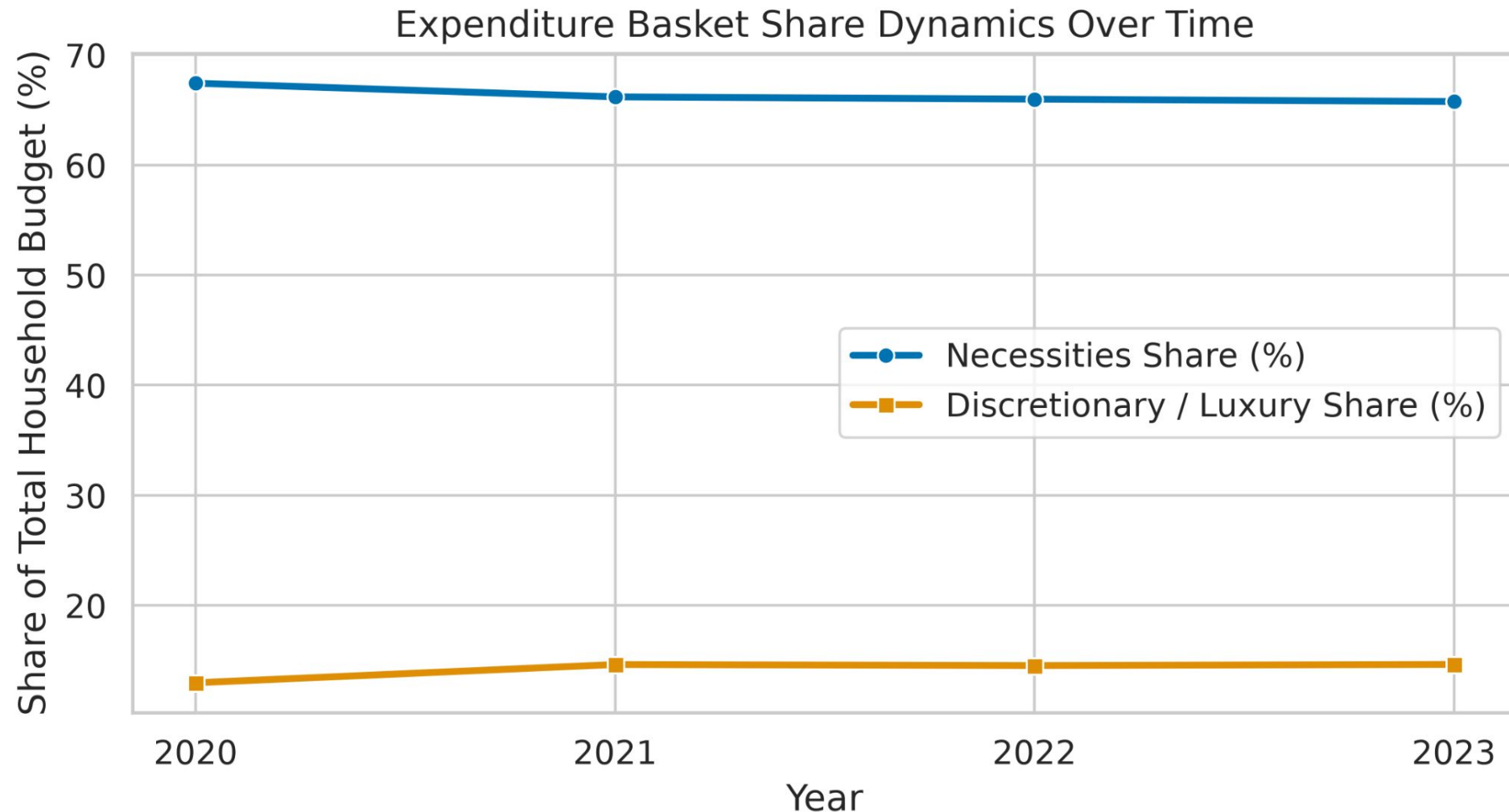
Workflow:

1. **Data Load:** Cleaned and parsed the 2020–2023 retail basket dataset ([consumer_basket_2020_2023.csv](#)).
2. **Feature Prep:** Computed monthly inflation changes and percentage shares for each product category.
3. **Threshold Analysis:** Filtered data around key inflation points to spot trend shifts ([01_inflation_threshold.png](#)).



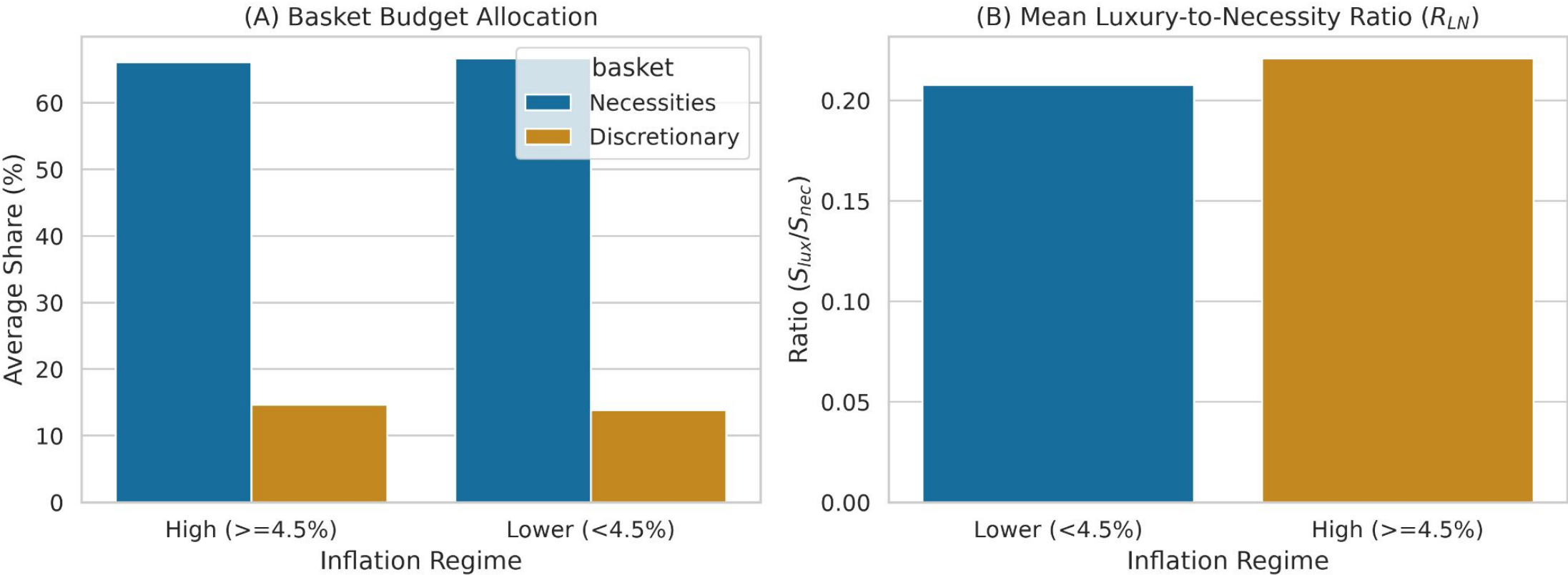
01_inflation_threshold.png

4. **Visuals:** Built share distribution plots over time and comparison charts (02_basket_shares_over_time.png,



02_basket_shares_over_time.png

Comparative Analysis Across Inflation Regimes



03_threshold_comparison.png

FEASIBILITY AND VIABILITY

- **Feasibility Analysis**

The code is lightweight, runs fast, and can plug into standard store checkout logs or ERP exports.

- **Challenges & Risks**

1. Missing granular socio-demographic labels.
2. Supply chain distortions (e.g., stockouts) confounding pure inflation effects.

- **Mitigation Strategies**

1. Applied rolling averages to smooth out sudden supply spikes.
2. Structured the script so live economic APIs can be plugged in later.

IMPACT AND BENEFITS

- **Impact on Target Audience**

1. Enables retailers to optimize inventory, pricing, and dynamic promotional discounting before margin loss occurs.
2. Helps policymakers understand the true living-cost burden on essential goods versus discretionary spend.

- **Key Benefits**

1. **Economic:** Minimizes deadweight loss and prevents overstocking of discretionary goods during inflationary spikes.
2. **Operational:** Provides clear quantitative triggers to adjust supply chain procurement schedules.

RESEARCH AND REFERENCES

1. U.S. Bureau of Labor Statistics, "Consumer Expenditures in 2023," BLS Report Series, 2024. [Online].
Available:
[https://www.bls.gov/opub/reports/consumer-expenditures/2023/home
.htm](https://www.bls.gov/opub/reports/consumer-expenditures/2023/home.htm)
2. U.S. Bureau of Labor Statistics, "Consumer Price Index for All Urban Consumers (CPI-U)," Data Series, 2024. [Online].
Available: <https://www.bls.gov/cpi/data.htm>
3. World Bank, "World Development Indicators: Inflation, Consumer Prices," World Bank Open Data, 2024.